

Ex No: 2 Chebyshev IIR Filter

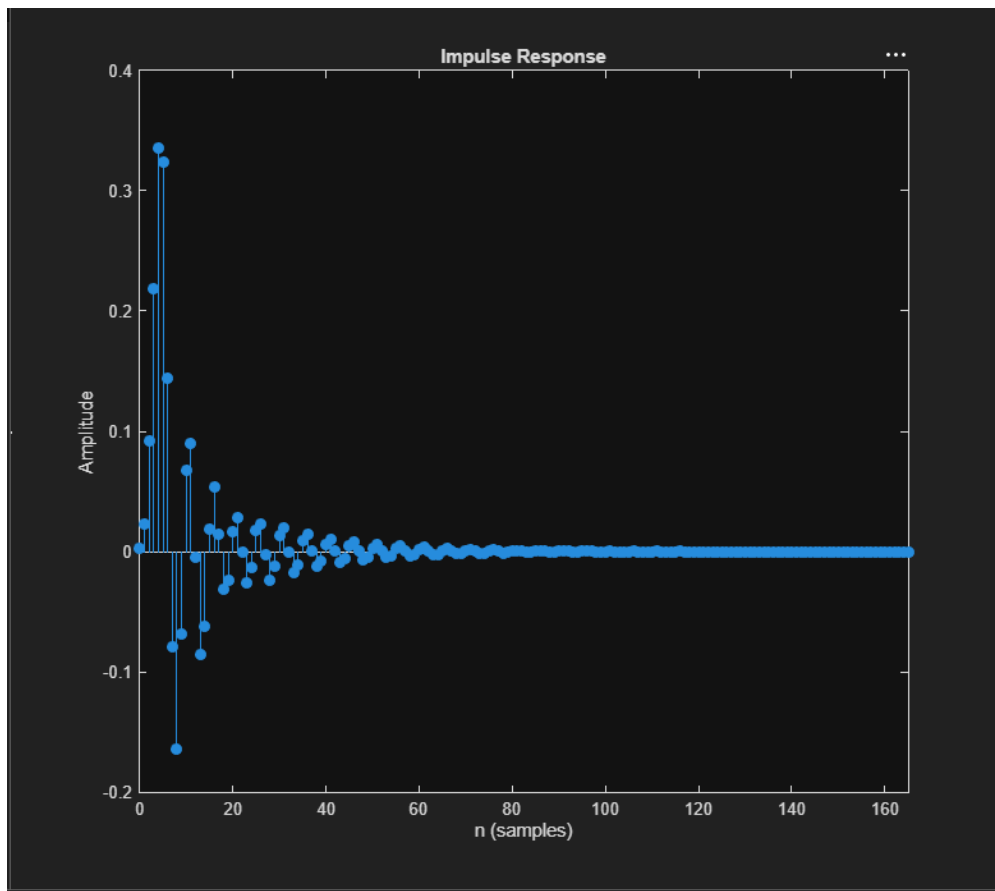
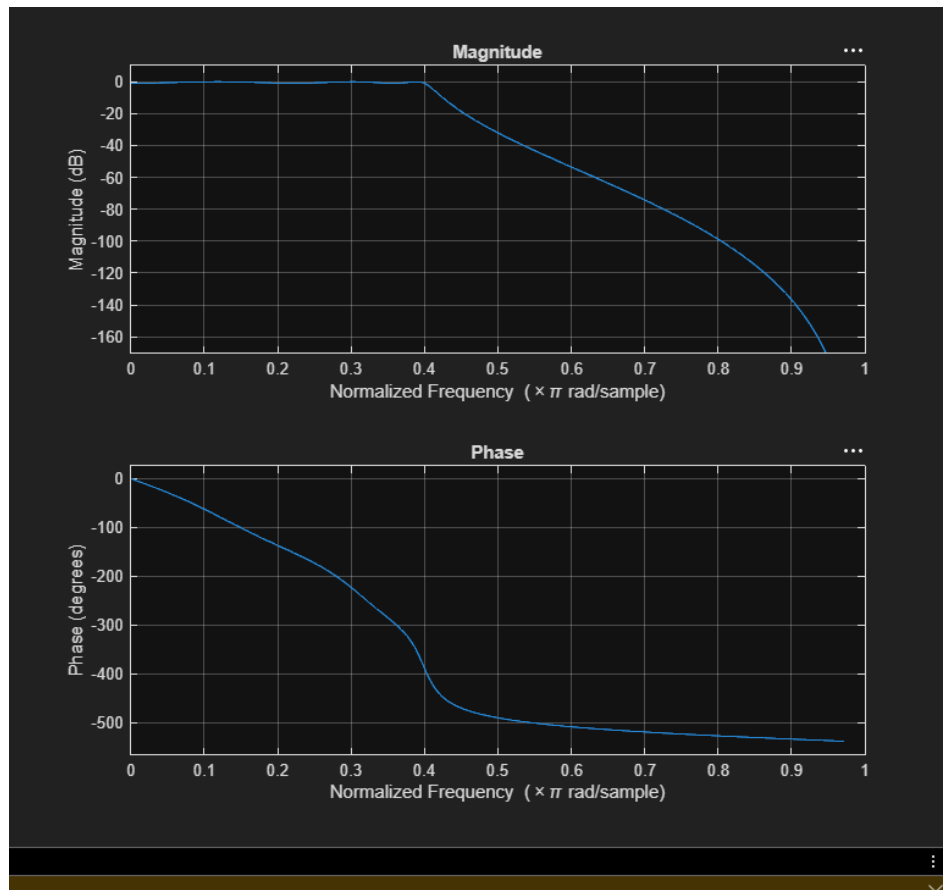
2.1 Design and Performance Evaluation of a Chebyshev Type-I IIR Low-Pass Filter using MATLAB

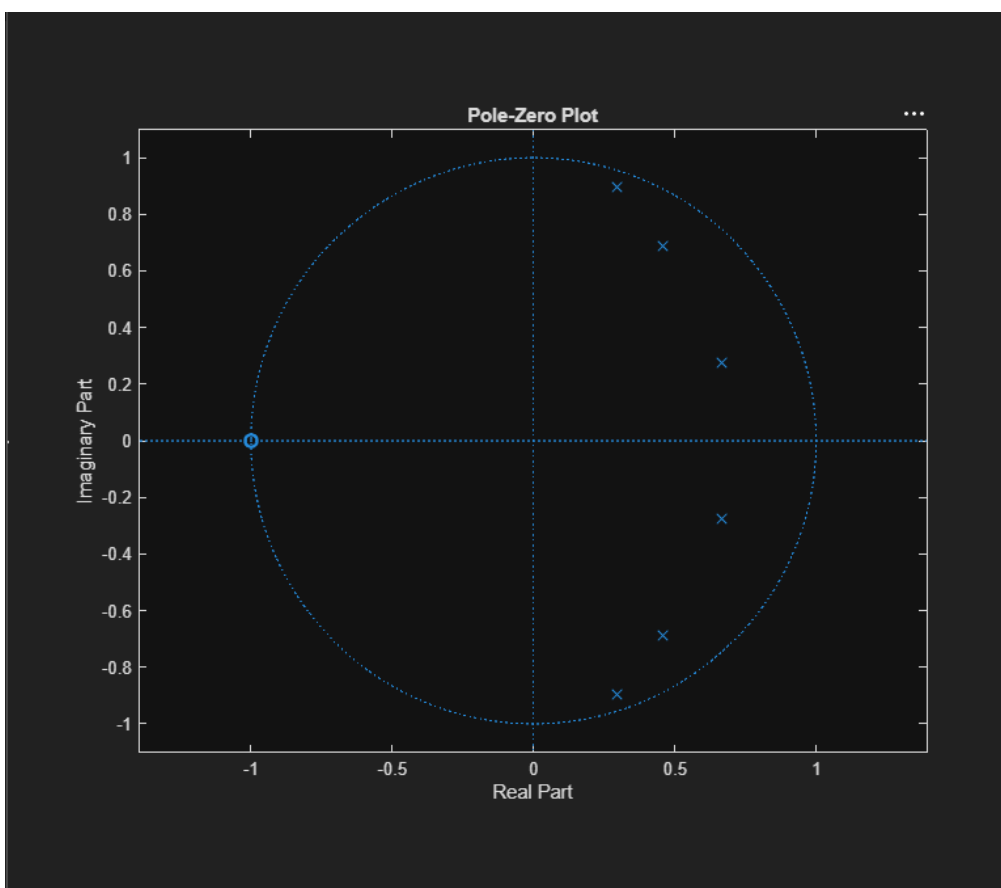
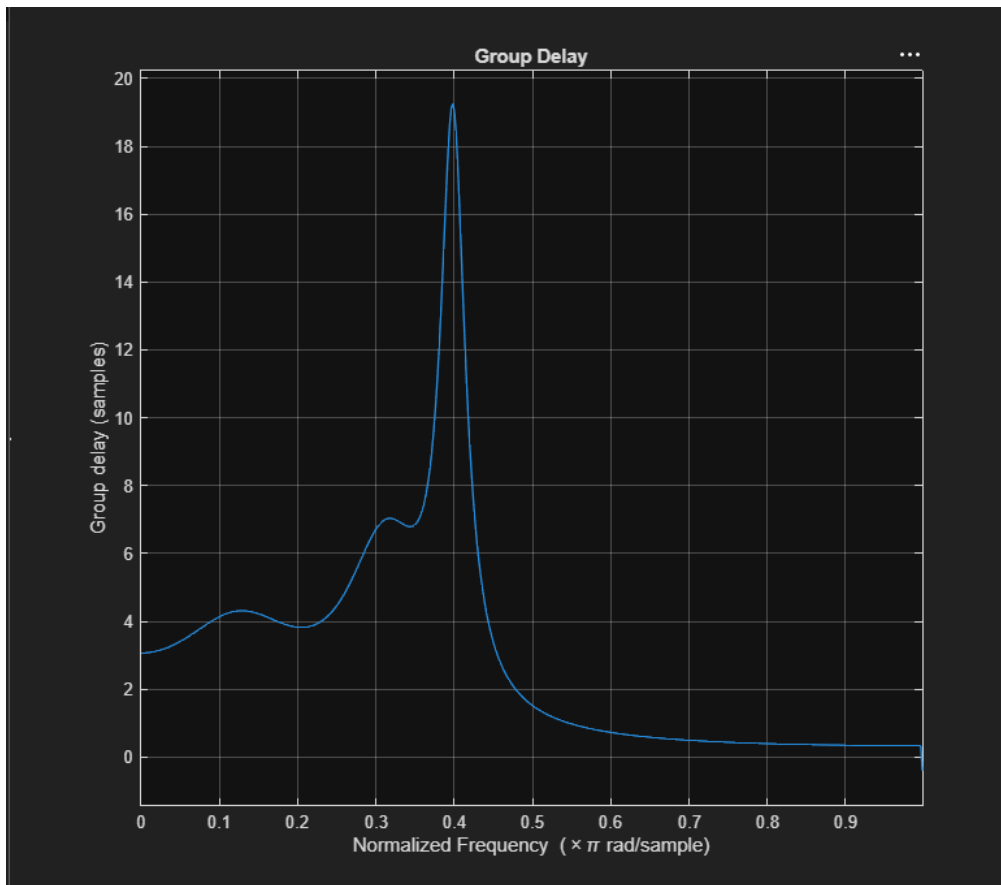
```
1  clc; clear;
2
3  Wp=0.4; Ws=0.55; Rp=1; As=40;
4
5
6  [N,Wn]=cheb1ord(Wp,Ws,Rp,As);
7  [b,a]=cheby1(N,Rp,Wn);
8
9  fprintf('Filter Order = %d\n',N);
10 fprintf('Cutoff Frequency = %.2f * pi rad/sample\n',Wn);
11
12 if all(abs(roots(a))<1)
13     disp('System is STABLE');
14 else
15     disp('System is UNSTABLE');
16 end
17
18 % ----- GRAPHS -----
19 figure; freqz(b,a); % Magnitude + Phase
20 figure; impz(b,a); % Impulse Response
21 figure; grpdelay(b,a); % Group Delay
22 figure; zplane(b,a); % Pole-Zero
```

Command Window

New to MATLAB? See resources for [Getting Started](#).

Filter Order = 6
Cutoff Frequency = 0.40 * pi rad/sample
System is STABLE
>>





2.2 Design and Performance Evaluation of a Chebyshev Type-I IIR High-Pass Filter

```
1  clc;
2  clear;
3  close all;
4
5  % Specifications
6  Rp = 1; % Passband Ripple (dB)
7  Rs = 40; % Stopband Attenuation (dB)
8  Wp = 0.5; % Passband Edge
9
10
11
12  Ws = 0.35; % Stopband Edge
13
14  % Find Filter Order
15  [n,Wn] = cheblord(Wp,Ws,Rp,Rs);
16
17  % Design High-Pass Filter
18  [b,a] = cheby1(n,Rp,Wn,'high');
19
20  % Frequency Response
21  figure;
22  freqz(b,a);
23  title('Frequency Response');
24
25  % Impulse Response
26  figure;
27  impz(b,a);
28  title('Impulse Response');
29
30  % Group Delay
31  figure;
32  grpdelay(b,a);
33  title('Group Delay');
34
35  % Pole-Zero Plot
36  figure;
37  zplane(b,a);
38  title('Pole-Zero Plot');
39
40  fprintf('Filter Order = %d\n',n);
41
42  fprintf('Cutoff Frequency = %.4f\n',Wn);
43
44  Command Window
45
46  New to MATLAB? See resources for Getting Started.
47
48  Filter Order = 6
49  Cutoff Frequency = 0.5000
50  >>
```

